Furuta Pendulum RL — Project Timeline

From a level-ground balancer to two-axis tilt — methods, challenges, results

PROJECT 1 · Level-ground Furuta

deployed, working on hardware

Goal **Level-ground Furuta on GM3506 gimbal**

Classical LQR balances in sim but FAILS on hardware — pivot friction winds the arm to the ±180° cable limit (no integral action vs the friction offset).

Method **One TQC policy for swing-up AND balance**

MuJoCo + wide domain randomization + curriculum. Obs 6-D [cosθ, sinθ, θ̇, φ, φ̇, prev_action]; reward = cos(θ) backbone + CAPS action-smoothness; bonus/success gated on arm < 90° (kills the 'balance at the cable edge' loophole).

Findings **Realism fears were mostly pessimistic**

Actuator step-test: real lag ~15 ms, already modeled → swing-up transfers. Ablation flagged 'motor authority matters' (KM×0.75 → 17% success) — a precursor to the 2D 11 V result.

2026-06-24 **RESULT: deployed on-chip, works on hardware**

Sim-trained TQC runs standalone on the ESP32 (MODE_RL) doing the full task; ~88% under randomized DR. This is the 'trained from nothing' origin.

PROJECT 2 · One-axis tilting board (1D)

verified 91.5% master

Goal **Board tilts ±20–30° about ONE axis**

Keep balancing to TRUE gravity-vertical through the motion. Add a BNO086 IMU (board tilt β, β̇) → 8-D obs; retarget reward from base-frame 'up' to true vertical.

Phase 0 **Sim feasibility confirmed**

±30° is achievable but orientation-dependent (arm at φ≈90° is the hard case where tilt most drives the pole).

2026-06-26 **Challenge: entropy collapse**

Stage-0 learned then oscillated/dropped — ent_coef ran away under gSDE + auto target-entropy. Fix: turn gSDE off. Also learned: use nenv=8, not 16 (better sample efficiency here).

06-27→29 **Phase C: metric fix + retention-aware fine-tuning**

Corrected success = balanced ≥80% of the final 2 s (not just a brief catch). Framed fine-tuning as a forgetting problem → teacher replay + behavior-cloning (RetentionTQC).

2026-06-29 **Verified master: 91.5%**

Clean ±20° free-arm master (clean20_master_verified91p5) — 915/1000 sustained. The anchor model for all later work.

2026-06-29 **Key finding: action DELAY is the DR bottleneck**

Ablation: mechanical/sensor DR barely matters; delay does (2-step → ~40%, 3-step → ~0%). It's non-Markov — a bounded residual can't fix an unobserved phase lag. Residual RL preserved but did NOT beat the clean master.

PROJECT 3 · Two-axis board (2D)

2026-06-30 — in progress

Setup **Two-axis (roll + pitch) board, 10-D obs**

Nested-gimbal model + continuous motion generator + BNO086 model. Training-free 1D→2D warm-start (transferred master, verified equivalent to 2e-6).

Step 4 **Warm-start: great everywhere except fast pitch**

~99% on level / roll / slow / static, but ~10% on FAST continuous pitch. That single gap became the whole problem to solve.

Step 5 Diagnosis: it's the CRITIC, not capability

Probe showed fast pitch is LEARNABLE (torque headroom at failure). Real cause: the transferred critic diverges NEGATIVE (predicts −1800 for states worth +1000), starving the actor.

Phase A/B Fix: reset the critic + warmup + γ=0.99

Discard the transferred critic, re-initialize it, warm it up with the actor frozen until calibrated, lower γ to 0.99. Divergence gone; co-training clears the ±10° regimes.

Step 6 Envelope decision: ±10°, up to 120°/s

±15°-fast is a REAL physics wall at 6 V (servo test confirmed it's not a sim artifact) → scoped the deployment target to ±10°.

Scale 5 seeds parallel on the server

5× RTX 6000 Ada. Verified local seed 0 (500 ep): ~100% slow/mid, ~34% at both ±10° 120°/s, with a calibrated critic. En route: 4/5 seeds NaN-crashed → added gradient clipping.

2026-06-30 11 V discovery: authority erases the wall

Motor runs ~11 V (sim used a conservative 6 V ≈ 1.83× less torque). Retrained at 11 V (teacher-free) → cleared the ENTIRE ±15° ladder at 0.93–0.97 (vs 0.10–0.33 at 6 V). Authority, not physics, was the limit. Pending 500-ep verify + a hardware thermal check.

SUCCESSFUL METHODS *what worked, and why*

Transfer, don't restart

Warm-start each project from the previous project's VERIFIED model instead of training from scratch — preserves proven swing-up/balance and avoids TQC seed-variance failures.

TQC + CAPS + true-vertical reward

Truncated Quantile Critics (off-policy, tiny deployable actor); CAPS action-smoothness so control transfers to a real motor; reward on TRUE gravity-vertical once the base tilts.

Critic reset + frozen-actor warmup + γ=0.99

The key 2D fix. Discard the out-of-distribution transferred critic, re-initialize it, and calibrate it with the actor FROZEN before it drives the actor; lower γ for stable, lower-variance targets. Cures the negative-divergence stall.

Retention fine-tuning (teacher + behavior-cloning)

Fine-tune without forgetting: mix successful old-policy transitions into replay and add an MSE loss pulling the actor toward them. Disable when the dynamics change (e.g. 11 V).

Soft, advancing curriculum

Ramp the disturbance in small steps; advance on a soft success threshold OR a timeout that ADVANCES (never kills) the seed — avoids the brittle-gate trap that stalled early runs.

Gradient clipping

Clip the gradient norm to stop the exploding-gradient NaN divergence — essential with a re-initialized critic (it crashed 4/5 seeds before we added it).

Verify before you trust

Select on SUSTAINED success (balanced ≥80% of the final 2 s), over ≥500 fresh episodes with Wilson 95% CIs; keep best-per-stage checkpoints, never the final one. Small peaks are noise.

Cheap diagnostic probes

Targeted probes — Q vs return-to-go calibration, action-saturation at failure — separate a training bug from a physics/authority limit BEFORE spending GPU hours.

Reproducible on-chip deployment

Export the actor MLP to a C header, verify NumPy inference vs SB3 to <1e-6 on CPU, and rebuild the identical observation in firmware at 200 Hz. This is what put P1 on hardware.

KEY LESSONS

- Transfer, don't restart — every project warm-started from the previous verified model.
- Diagnose with cheap probes before spending compute.
- Capability vs training: 'not saturated at failure' = trainable; 'authority ablation kills it' = physics/hardware.
- Trust 500-episode verified numbers + Wilson CIs — never a small training-eval peak.
- Instrument the mechanism: Q-vs-return, retention-loss magnitude, log-probs — bugs are invisible in success rate alone.

NOW: 11 V verification + five 6 V baseline seeds training on the server. Current best deployable model: local 2D seed 0 (verified). Full detail: PROJECT_LINEAGE_METHODS_AND_LESSONS.md.